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HOME PLOTS APPS FIGURE FORMAT TOOLS Search (Ctrl+Shift+Space) Rabiya

Save As | Show Code | Title | Subtitle | X-Label | Y-Label | Z-Label | Legend | Colorbar | Grid | X-Grid | Y-Grid

FILE | LABELS AND ANNOTATIONS

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Editor

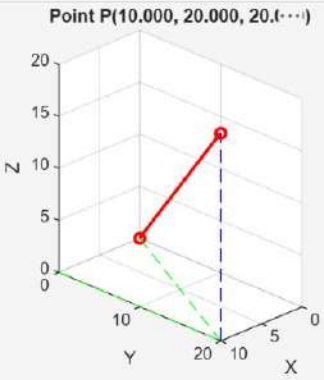
untitled.m

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```
1   clc;
2   clear;
3   close all;
4   x = input('Enter the value of X = ');
5   y = input('Enter the value of y = ');
6   z = input('Enter the value of z = ');
7   rho = sqrt(x^2 + y^2); phi = atan2d(y,x);
8   disp('Cylindrical coordinates [rho phi z] = '); disp([rho phi z]);
9   r = sqrt(x^2 + y^2 + z^2); theta = acosd(z/r);
10  disp('Spherical coordinates [r theta phi] = '); disp([r theta phi]).
11  % ---- plotting code ----
12  figure; plot3([0 x],[0 y],[0 z],'r-o','LineWidth',2); hold on;
13  plot3([x x],[y y],[0 z],'b--'); plot3([0 x],[0 y],[0 0],'g--'); plot
14  grid on; axis equal; xlabel('X'); ylabel('Y'); zlabel('Z');
15  title(sprintf('Point P(%.3f, %.3f, %.3f)',x,y,z)); view(135,25);
16
```

Figures

Figure 1



Command Window

```
Spherical coordinates [r theta phi] =
   30.0000   48.1897   63.4349

>>
```

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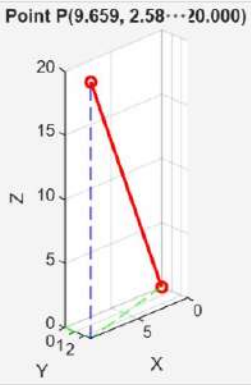
untitled.m

/MATLAB Drive/untitled.m

```
clc;
clear;
close all;
rho = input('Enter the value of rho = ');
phi = input('Enter the value of phi = ');
z = input('Enter the value of z = ');
x = rho*cosd(phi); y = rho*sind(phi);
disp('Cartesian coordinates [x y z] = '); disp([x y z]);
r = sqrt(rho^2 + z^2); theta = acosd(z/r);
disp('Spherical coordinates [r theta phi] = '); disp([r theta phi]).
% ---- plotting code ----
figure; plot3([0 x],[0 y],[0 z],'r-o','LineWidth',2); hold on;
plot3([x x],[y y],[0 z],'b--'); plot3([0 x],[0 y],[0 0],'g--'); plot
grid on; axis equal; xlabel('X'); ylabel('Y'); zlabel('Z');
title(sprintf('Point P(%.3f, %.3f, %.3f)',x,y,z)); view(135,25);
```

Figures

Figure 1



Command Window

```
Enter the value of z = 20
Cartesian coordinates [x y z] =
    9.6593    2.5882   20.0000

Spherical coordinates [r theta phi] =
   22.3607   26.5651   15.0000
```

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Save As | Show Code | Title | Subtitle | X-Label | Y-Label | Z-Label | Legend | Colorbar | Grid | X-Grid | Y-Grid

FILE | LABELS AND ANNOTATIONS

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Editor

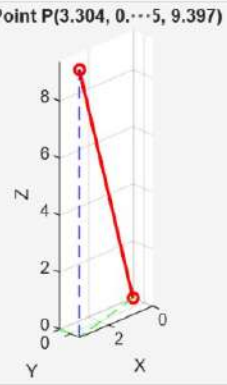
untitled.m

/MATLAB Drive/untitled.m

```
1   clc;
2   clear;
3   close all;
4   r = input('Enter the value of r = ');
5   phi = input('Enter the value of phi = ');
6   theta = input('Enter the value of theta = ');
7   x = r*sind(theta)*cosd(phi); y = r*sind(theta)*sind(phi); z = r*cosd
8   disp('Cartesian coordinates [x y z] = '); disp([x y z]);
9   rho = r*sind(theta);
10  disp('Cylindrical coordinates [rho phi z] = '); disp([rho phi z]);
11  % ---- plotting code ----
12  figure; plot3([0 x],[0 y],[0 z],'r-o','LineWidth',2); hold on;
13  plot3([x x],[y y],[0 z],'b--'); plot3([0 x],[0 y],[0 0],'g--'); plot
14  grid on; axis equal; xlabel('X'); ylabel('Y'); zlabel('Z');
15  title(sprintf('Point P(%.3f, %.3f, %.3f)',x,y,z)); view(135,25);
16
```

Figures

Figure 1



Command Window

```
Enter the value of theta = 20
Cartesian coordinates [x y z] =
    3.3037    0.8852    9.3969

Cylindrical coordinates [rho phi z] =
    3.4202   15.0000    9.3969
```

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